

Question

Chromium is a steel-gray metal extracted from red Siberian ore by the French scientist Vauquelin in 1797. It is often used in the manufacture of stainless steel and for chrome plating metal surfaces of instruments. Industrially, the preparation of elemental chromium starts from chromite (FeCr_2O_4), which is strongly heated in air with a mixture of sodium carbonate, where iron is converted to ferric oxide (Reaction 1). The resulting solid residue is leached with water, acidified, and a chromium-containing sodium salt precipitates. This salt is then reduced to chromium oxide by heating with carbon at high temperature, simultaneously generating a salt and releasing a gas (Reaction 2). Finally, treatment with ignited metallic aluminum yields relatively pure chromium (Reaction 3).

For the extraction of molybdenum, which is in the same group as chromium, molybdenite is usually used as the raw material, which is roasted in oxygen (Reaction 4). The resulting product is leached with ammonia water to transfer molybdenum to the solution; alternatively, molybdenite can react with caustic soda solution in an oxygen atmosphere to transfer molybdenum to the solution (Reaction 5), during which sulfur is also oxidized and enters the solution. Subsequently, the ammonia-leached solution is acidified to precipitate a polyoxometalate which anion contains seven molybdenum atoms (Reaction 6, ionic equation), which is then calcined and reduced with hydrogen to obtain relatively pure metallic molybdenum.

Let the sum of the coefficients of reactants and products in the equations of Reactions 1 to 6 (requiring the coefficient ratio to be the simplest integer ratio) be $a_i, i = 1, 2, 3, 4, 5, 6$, and the absolute value of the difference between the coefficients of reactants and products be $b_i, i = 1, 2, 3, 4, 5, 6$, then the value of $\log \frac{E(a)}{D(a)} \frac{E(b)}{D(b)}$ is (retain two decimal places in the calculation process, and the final result error within 0.01 is considered correct; Here, D is calculated using the population standard deviation):

A. All other options are incorrect

B. -0.98

C. -0.83

D. -0.69

E. -0.52

F. -0.35

G. -0.12

H. 0.17

I. 0.33

J. 0.64

K. 0.81

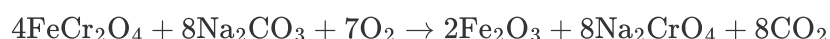
L. 1.02

Answer

Correct Answer: **F**

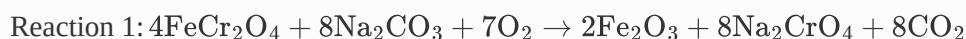
Detailed Explanation

The solid obtained after reaction 1 can be leached with water to obtain a chromium-containing sodium salt. Therefore, reaction 1 should produce +6 valent chromium. Thus, reaction 1 is:

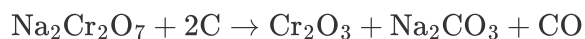


CHECKPOINT

1 PTS

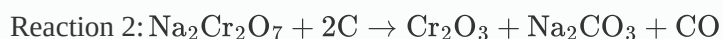


After leaching the product with water and acidifying it, sodium chromate is converted to sodium dichromate. Sodium dichromate is reduced to chromium oxide, which should be chromium(III) oxide, when co-heated with carbon at high temperature, and the gas should be carbon monoxide. Therefore, reaction 2 is:

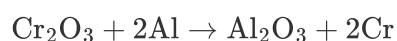


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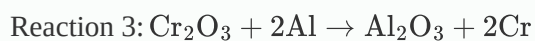


The final step is a displacement reaction involving aluminum, and the equation is:

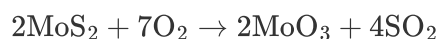


CHECKPOINT

1 PTS



The main component of molybdenite is MoS_2 , and the equation for its roasting in oxygen is:



CHECKPOINT

1 PTS



When molybdenite reacts with sodium hydroxide solution in an oxygen atmosphere, the sulfur element can be oxidized to +6 valence. Therefore, the equation is:

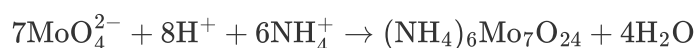


CHECKPOINT

1 PTS



Reaction 6 produces a polymolybdate containing 7 molybdenum atoms, and its equation is:



CHECKPOINT

1 PTS



Therefore:

1	2	3	4	5	6	
a	37	6	6	15	35	26
b	1	0	0	3	11	16

$$E(a) = 20.83, D(a) = 12.67, E(b) = 5.17, D(b) = 6.15$$

$$\frac{E(a)}{D(a)} = 1.64, \frac{E(b)}{D(b)} = 0.84$$

CHECKPOINT

0.5 PTS

$$\frac{E(a)}{D(a)} = 1.64$$

CHECKPOINT

0.5 PTS

$$\frac{E(b)}{D(b)} = 0.84$$

$$\log \frac{E(a)}{D(a)} \frac{E(b)}{D(b)} = -0.35$$